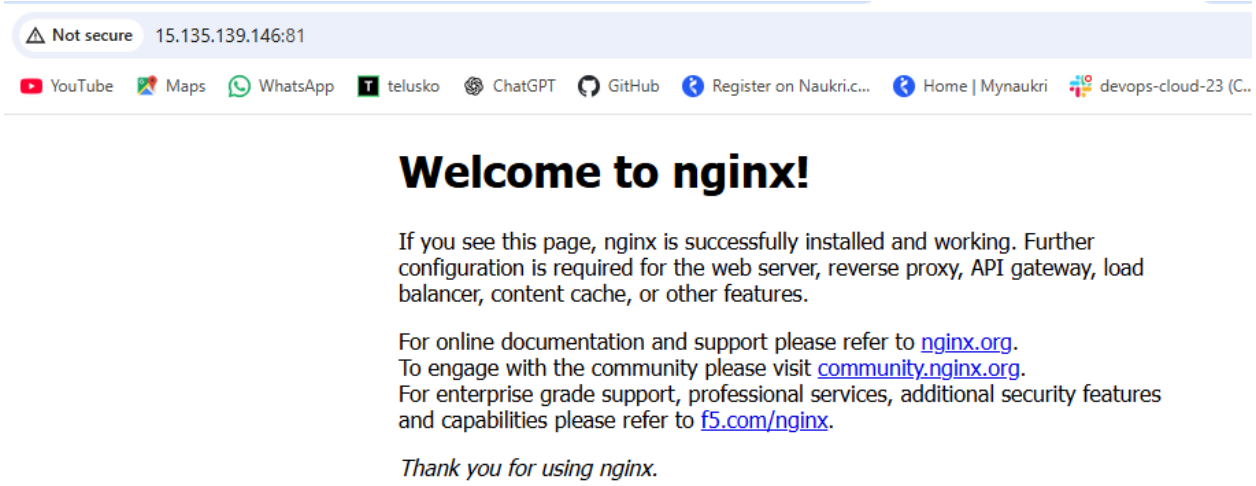
1.Install nginx and run nginx on port number 81.

```
[ec2-user@ip-172-31-22-105 ~]$ sudo systemctl restart nginx
[ec2-user@ip-172-31-22-105 ~]$ sudo systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
     Active: active (running) since Wed 2026-10-14 10:14:43 UTC; 23s ago
    Process: 125314 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
    Process: 125322 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
    Process: 125328 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
   Main PID: 125330 (nginx)
      Tasks: 3 (limit: 1059)
     Memory: 3.5M
        CPU: 62ms
     CGroup: /system.slice/nginx.service
             ├─125330 "nginx: master process /usr/sbin/nginx"
             ├─125331 "nginx: worker process"
             └─125332 "nginx: worker process"

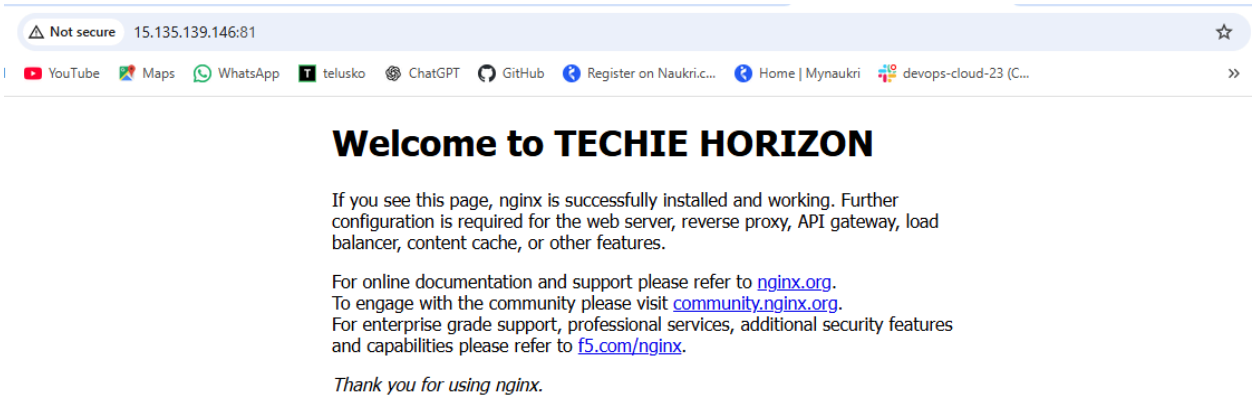
Oct 14 10:14:43 ip-172-31-22-105.ap-southeast-2.compute.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server.
Oct 14 10:14:43 ip-172-31-22-105.ap-southeast-2.compute.internal nginx[125322]: nginx: the configuration file /etc/nginx/nginx.conf syntax is
Oct 14 10:14:43 ip-172-31-22-105.ap-southeast-2.compute.internal nginx[125322]: nginx: configuration file /etc/nginx/nginx.conf test is succes
Oct 14 10:14:43 ip-172-31-22-105.ap-southeast-2.compute.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
[ec2-user@ip-172-31-22-105 ~]$ sudo ss -lntp | grep ':81'
LISTEN 0      511          0.0.0.0:81        0.0.0.0:*    users:(("nginx",pid=125332,fd=8),("nginx",pid=125331,fd=8),("nginx",pid=125330,
LISTEN 0      511             [::]:81           [::]:*    users:(("nginx",pid=125332,fd=9),("nginx",pid=125331,fd=9),("nginx",pid=125330,
[ec2-user@ip-172-31-22-105 ~]$ curl -I http://localhost:81
HTTP/1.1 200 OK
Server: nginx/1.30.5
Date: Wed, 14 Oct 2026 10:17:49 GMT
Content-Type: text/html
Content-Length: 896
```



1. I updated the Amazon Linux packages.
2. I installed Nginx on the server.
3. I started the Nginx service.
4. I checked the Nginx service status to make sure it was running.
5. I opened the Nginx configuration file.
6. I changed the default Nginx listening port from 80 to 81.
7. I saved the Nginx configuration file.
8. I tested the Nginx configuration to make sure there were no errors.

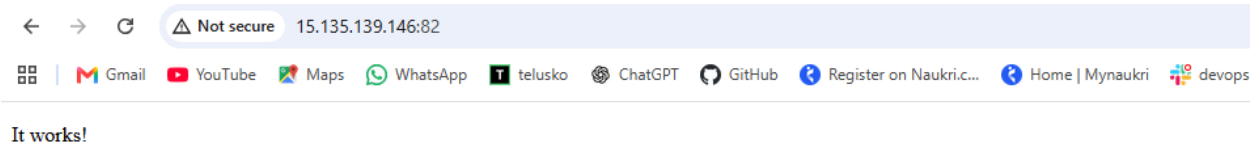
9. I restarted Nginx to apply the new port configuration.
10. I verified that Nginx was listening on port 81.
11. I allowed TCP port 81 in the AWS EC2 Security Group.
12. I accessed Nginx from the browser using the EC2 public IP and port 81.
13. I verified that the Nginx welcome page was displayed successfully.

2.Deploy a sample index.html file on nginx.



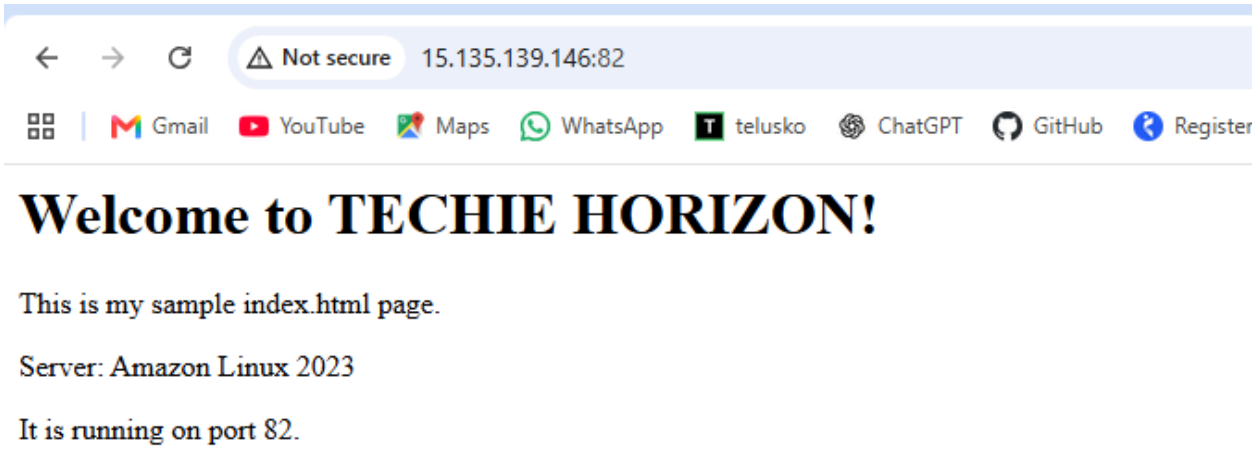
1. I navigated to the Nginx web root directory.
2. I created a new index.html file.
3. I added sample HTML content to the index.html file.
4. I saved the index.html file in the Nginx web directory.
5. I verified that the index.html file was created successfully.
6. I tested the webpage locally using the Nginx port.
7. I accessed the webpage using the EC2 public IP address and port 81.
8. I verified that my sample index.html page was displayed successfully in the browser.

3.Install Apache and run Apache on port number 82



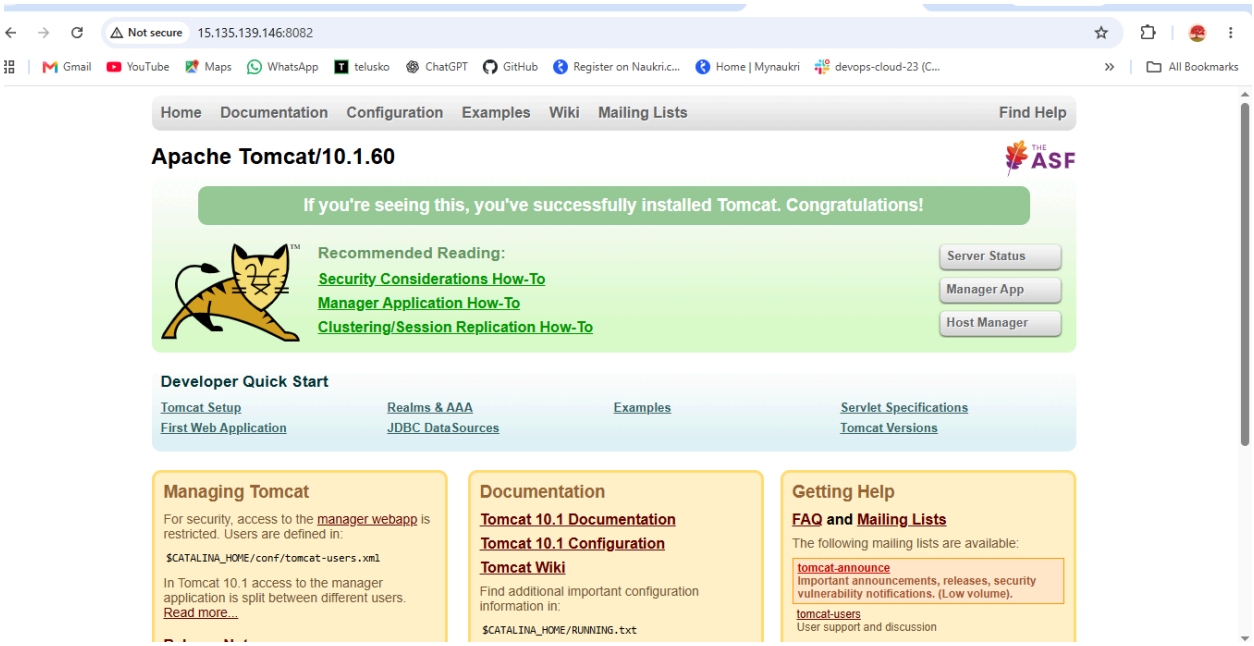
1. I installed the Apache HTTP Server on the Linux server.
2. I opened the Apache configuration file.
3. I changed the default Apache port from 80 to 82.
4. I checked the Apache configuration for any syntax errors.
5. I restarted the Apache service to apply the port change.
6. I verified that Apache was running successfully on port 82.
7. I tested the Apache server locally using port 82.
8. I accessed Apache using the EC2 public IP address and port 82 in the browser.
9. I verified that the Apache default webpage was displayed successfully.

4.Deploy a sample index.html file on Apache.



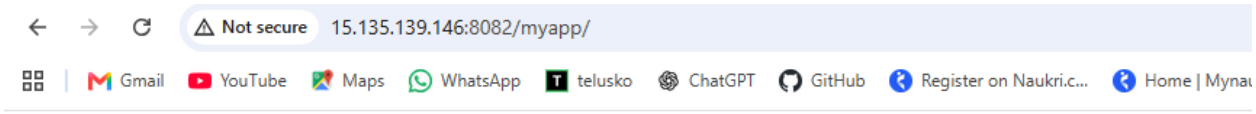
1. I navigated to the Apache web root directory.
2. I created a new index.html file.
3. I added sample HTML content to the index.html file.
4. I saved the index.html file in the Apache web directory.
5. I verified that the index.html file was created successfully.
6. I tested the webpage locally using Apache on port 82.
7. I accessed the webpage using the EC2 public IP address and port 82.
8. I verified that my sample index.html page was displayed successfully in the browser.

5.Install Apache tomcat on port number 8082



1.I installed Apache Tomcat 10 on the Linux server.
2.I opened the Tomcat configuration file.
3.I changed the default Tomcat port from 8080 to 8082.
4.I saved the Tomcat configuration after changing the port.
5.I started the Tomcat service.
6.I checked the Tomcat service status to make sure it was running.
7.I verified that Tomcat was listening on port 8082.
8.I tested the Tomcat server locally using port 8082.
9.I accessed Tomcat using the EC2 public IP address and port 8082 in the browser.
10.I verified that Tomcat was running successfully on port 8082.

6.Deploy a sample app on webapps



Welcome to My Sample Web Application!

My application is running on Amazon Linux 2023.

Server: Apache Tomcat

Port: 8082

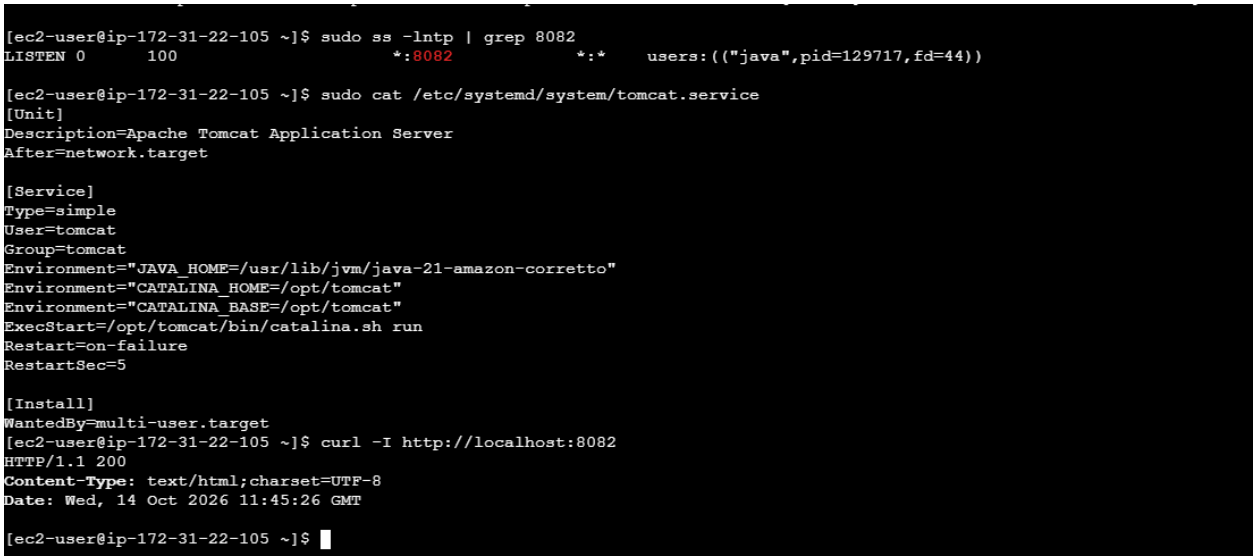
1.I navigated to the Tomcat webapps directory.
2.I created a new folder for my sample application.
3.I created a sample index.html file inside the application folder.
4.I added sample content to the application.
5.I restarted the Tomcat service.
6.I tested the sample application using port 8082.
7.I accessed the application using the EC2 public IP address and port 8082.
8.I verified that my sample application was displayed successfully.

7.Create a tomcat.service file for tomcat.

```
[ec2-user@ip-172-31-22-105 ~]$ sudo systemctl daemon-reload
[ec2-user@ip-172-31-22-105 ~]$ sudo systemctl status tomcat
● tomcat.service - Apache Tomcat Application Server
     Loaded: loaded (/etc/systemd/system/tomcat.service; enabled; preset: disabled)
     Active: active (running) since Wed 2026-10-14 11:19:19 UTC; 22min ago
   Main PID: 129717 (java)
      Tasks: 29 (limit: 1059)
     Memory: 96.5M
        CPU: 7.430s
     CGroup: /system.slice/tomcat.service
             └─129717 /usr/lib/jvm/java-21-amazon-corretto/bin/java -Djava.util.logging.config.file=/opt/tomcat/conf/logging.properties -Djava.util.l>

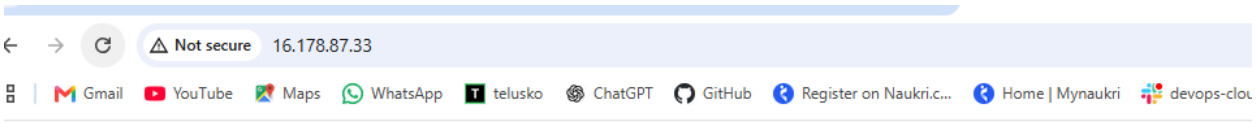
Oct 14 11:19:21 ip-172-31-22-105.ap-southeast-2.compute.internal catalina.sh[129717]: 14-Oct-2026 11:19:21.710 INFO [main] org.apache.catalina.startu>
Oct 14 11:19:22 ip-172-31-22-105.ap-southeast-2.compute.internal catalina.sh[129717]: 14-Oct-2026 11:19:22.040 INFO [main] org.apache.catalina.startu>
Oct 14 11:19:22 ip-172-31-22-105.ap-southeast-2.compute.internal catalina.sh[129717]: 14-Oct-2026 11:19:22.042 INFO [main] org.apache.catalina.startu>
Oct 14 11:19:22 ip-172-31-22-105.ap-southeast-2.compute.internal catalina.sh[129717]: 14-Oct-2026 11:19:22.092 INFO [main] org.apache.catalina.startu>
Oct 14 11:19:22 ip-172-31-22-105.ap-southeast-2.compute.internal catalina.sh[129717]: 14-Oct-2026 11:19:22.094 INFO [main] org.apache.catalina.startu>
Oct 14 11:19:22 ip-172-31-22-105.ap-southeast-2.compute.internal catalina.sh[129717]: 14-Oct-2026 11:19:22.129 INFO [main] org.apache.catalina.startu>
Oct 14 11:19:22 ip-172-31-22-105.ap-southeast-2.compute.internal catalina.sh[129717]: 14-Oct-2026 11:19:22.138 INFO [main] org.apache.coyote.Abstract>
Oct 14 11:19:22 ip-172-31-22-105.ap-southeast-2.compute.internal catalina.sh[129717]: 14-Oct-2026 11:19:22.153 INFO [main] org.apache.catalina.startu>
Oct 14 11:25:32 ip-172-31-22-105.ap-southeast-2.compute.internal catalina.sh[129717]: 14-Oct-2026 11:25:32.200 INFO [Catalina-utility-2] org.apache.c>
Oct 14 11:25:32 ip-172-31-22-105.ap-southeast-2.compute.internal catalina.sh[129717]: 14-Oct-2026 11:25:32.274 INFO [Catalina-utility-2] org.apache.c>

[ec2-user@ip-172-31-22-105 ~]$ sudo ss -lntp | grep 8082
LISTEN 0      100                *:8082            *:*    users:(("java",pid=129717,fd=44))
```



1.I created a tomcat.service file in the systemd directory.
2.I added the Tomcat service configuration to the file.
3.I configured the Tomcat user, group, Java environment, and Tomcat path.
4.I configured the service to start and stop Tomcat properly.
5.I saved the tomcat.service file.
6.I reloaded the systemd configuration.
7.I enabled the Tomcat service to start automatically after reboot.
8.I started the Tomcat service.
9.I checked the service status to verify that Tomcat was running successfully.

8.Configure HA Proxy server

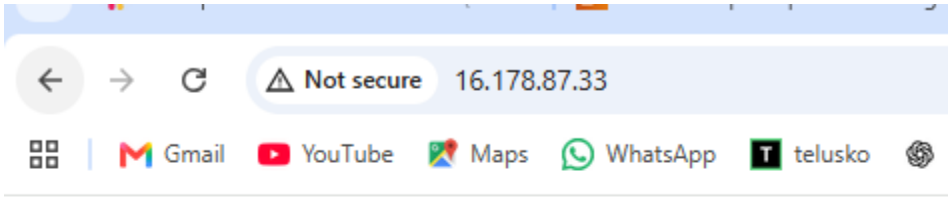


Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.

Thank you for using nginx.



It works!